C# (C-Sharp) Introduction (Detailed Guide for PDF)

What is C#?

C# is a modern, object-oriented programming language developed by Microsoft. It is mainly used with the .NET framework to build a wide range of applications.

History of C#

- Developed by Microsoft
- Designed by Anders Hejlsberg
- Released in 2000
- Part of .NET framework

Why Use C#?

- Easy to learn
- Strongly typed language
- Used for Windows, web, and game development
- Powerful libraries

Features of C#

- Object-Oriented Programming
- Type-safe
- Automatic memory management (Garbage Collection)
- Rich library support
- Cross-platform (.NET Core)

Basic Program Structure

```
using System;
```

```
class Program {
    static void Main(string[] args) {
        Console.WriteLine("Hello, World!");
    }
}
```

Variables

```
int x = 10;
string name = "John";
```

Data Types

- int
- float
- double
- char
- bool
- string

Operators

- Arithmetic (+, -, *, /)
- Relational (==, !=, >, <)
- Logical (&&, ||)

Conditional Statements

```
int age = 20;

if (age > 18) {
    Console.WriteLine("Adult");
} else {
    Console.WriteLine("Minor");
}
```

Loops

For Loop

```
for (int i = 0; i < 5; i++) {
    Console.WriteLine(i);
}
```

While Loop

```
int i = 0;
while (i < 5) {
    i++;
}
```

Methods (Functions)

```
static int Add(int a, int b) {
    return a + b;
}
```

Object-Oriented Programming

Class and Object

```
class Person {
    public string name;
}
```

Encapsulation

Inheritance

Polymorphism

Arrays

```
int[] numbers = {1, 2, 3};
```

Exception Handling

```
try {
    int x = 10 / 0;
} catch (Exception e) {
    Console.WriteLine("Error");
}
```

Collections

```
using System.Collections.Generic;

List<int> list = new List<int>();
list.Add(10);
```

File Handling

```
using System.IO;

File.WriteAllText("file.txt", "Hello");
```

Example Program

```
using System;

class Test {
    static void Main() {
        Console.Write("Enter name: ");
        string name = Console.ReadLine();
        Console.WriteLine("Hello " + name);
    }
}
```

Advantages of C#

- Easy to learn
- Secure
- Powerful framework support

Disadvantages of C#

- Mainly dependent on .NET
- Less control over hardware

Best Practices

- Use proper naming conventions
- Keep code clean
- Use exception handling